

Fluid-Structure Interaction and Differentiable

Optimization of Falling Leaves and Seeds

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Problem Statement



The descent of falling leaves is a classical fluid-structure interaction problem involving complex interactions among gravity, aerodynamic forces, body geometry, and inertia.

Rather than falling vertically, leaves often exhibit fluttering, tumbling, and lateral drifting, which influence transport distance and descent efficiency. This study investigates how internal mass distribution, represented by center-of-mass offset, affects the trajectory, stability, and transport performance of a falling leaf under crosswind conditions. The objective is to determine whether simple mass-distribution modifications can enhance lateral transport while maintaining stable descent, providing insights into natural dispersal mechanisms and the design of bio-inspired passive flying structures.

Literature Review

Previous studies on freely falling plates and leaf-like bodies have established the fundamental mechanisms of fluttering, tumbling, and steady descent through both experimental and numerical analyses [1]–[4]. Later work further demonstrated that physical parameters such as inertia, aspect ratio, and density ratio strongly influence descent behavior [5], [6]. More recently, Le et al. proposed a computational biomimetics framework for winged seeds, extending aerodynamic research from motion analysis toward bio-inspired design and optimization [7]. However, most existing studies focus on analyzing motion under prescribed conditions, while the role of internal mass distribution remains less explored. This motivates the present work, which investigates the effect of center-of-mass offset on the trajectory and transport performance of a falling leaf under crosswind conditions.

Methodology

• Problem Formulation:
We simulate the fluid-structure interaction (FSI) of a falling leaves within a 2D background wind field. The leaf is modeled as a oval with 3 degrees of freedom (2 translation, 1 rotation). By parameterizing the center-of-mass (CoM) location and aspect ratio, we investigate the aerodynamic instabilities, including fluttering and tumbling modes.

• Fluid Dynamics & Rigid Body Dynamics:
The background fluid domain is resolved via a Moment-Encoded Lattice Boltzmann Method:

$f_{t+1} = F_{lbm}(f_t, v_{bound})$

$F_{fluid} = G_{force}(f_t, v_{bound})$

The falling leaves kinematics is updated using Semi-Implicit Euler method:

$s_{t+1} = S_{rigid}(s_t, F_{fluid})$

$v_{bound} = K(s_t)$

• Differentiable Physics & Gradient Framework:
To maximize flight objectives (e.g. gliding range or loft time), we employ a fully-differentiable physics pipeline. Our system established a cross communication channel between Warp (fluid adjoint) with Pytorch (solid autograd), enabling automatic differentiation.
Gradient Backpropagation Pipeline:

$$\frac{\partial Loss}{\partial s_t} = \underbrace{\frac{\partial Loss}{\partial s_{t+1}} \frac{\partial S_{rigid}}{\partial s_t}}_{\text{Pytorch Graph}} + \underbrace{\frac{\partial Loss}{\partial F_{fluid}} \frac{\partial F_{fluid}}{\partial s_t}}_{\text{Pytorch Graph}} = \underbrace{\frac{\partial Loss}{\partial s_{t+1}} \frac{\partial S_{rigid}}{\partial s_t}}_{\text{Pytorch Graph}} + \underbrace{\frac{\partial Loss}{\partial s_{t+1}} \frac{\partial S_{rigid}}{\partial F_{fluid}} * \frac{\partial G_{force}}{\partial v_{bound}}}_{\text{Warp LBM Tape}} * \underbrace{\frac{\partial K}{\partial s_t}}_{\text{Kinematics}}$$

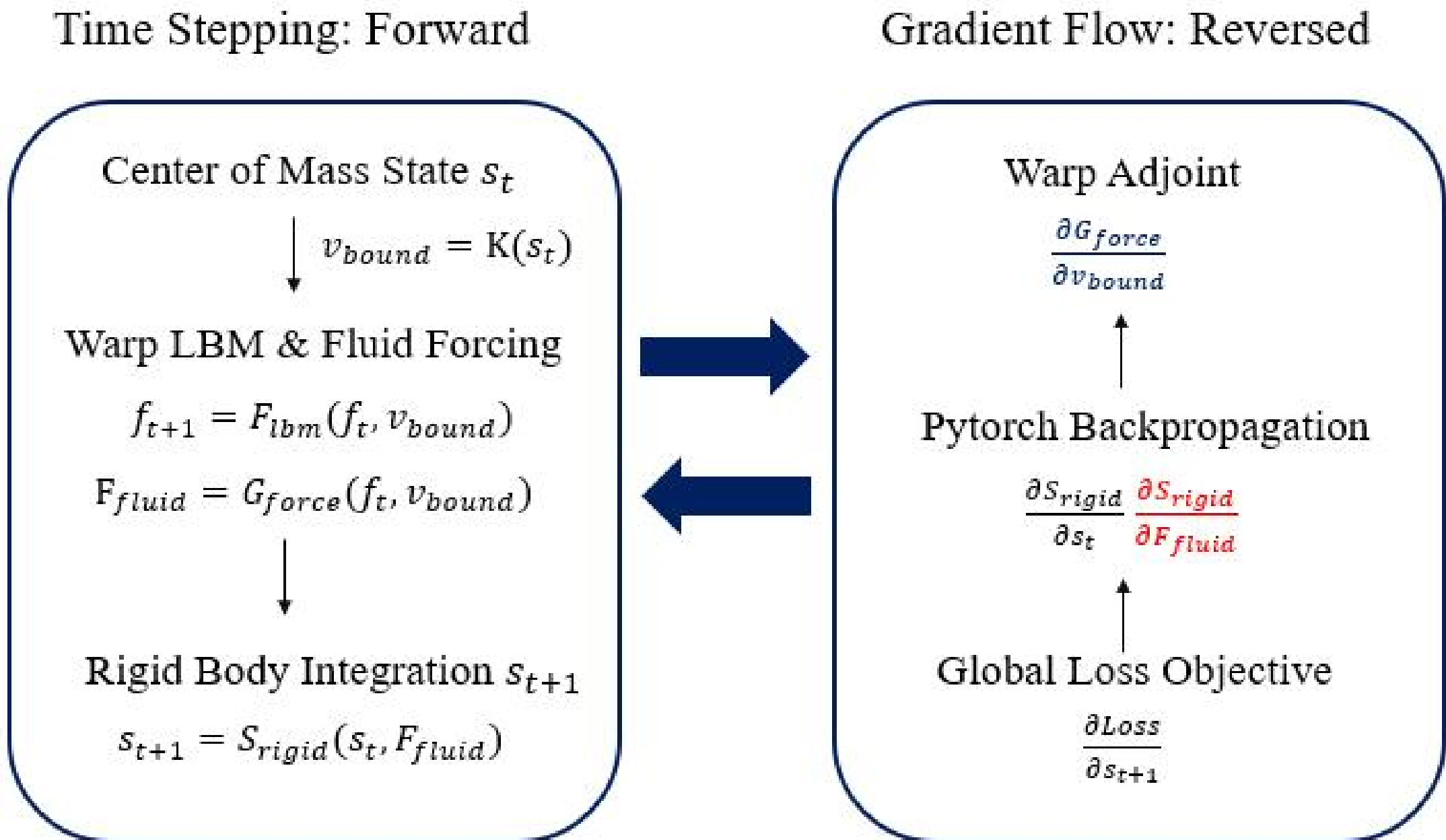


Figure 1 Computation and Backpropagation Architecture of the fully-differentiable FSI solver

Experimental Results

- Passive Descent Modes Across the Parametric Phase Space

As established by Andersen et al.[3], unconstrained falling plates exhibit rich passive dynamics. Over a sweeping parametric study across diverse aspect ratios (5.0, 6.0, 14.0), our framework demonstrates high numerical stability in capturing the transitions between stable tumbling, chaotic shedding, and periodic fluttering modes.

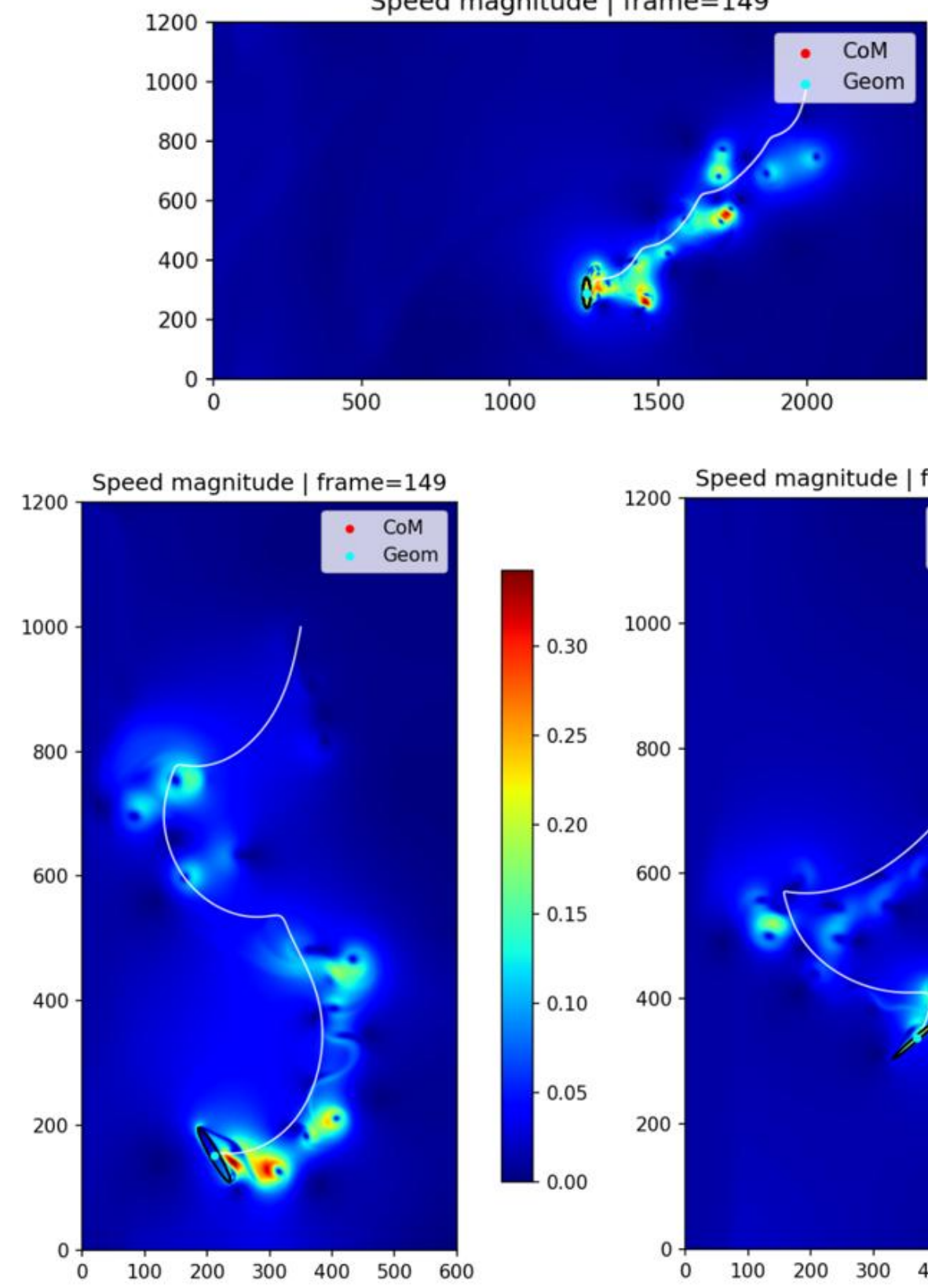


Figure 2 Hydrodynamic flow field snapshots of different passive descent modes

• Differentiable Optimization CoM Location
The optimization framework exhibits exceptional algorithmic robustness across varying physical configurations. These learned configurations effectively exploit aerodynamic forces to maximize both downstream transport distance and loft time.

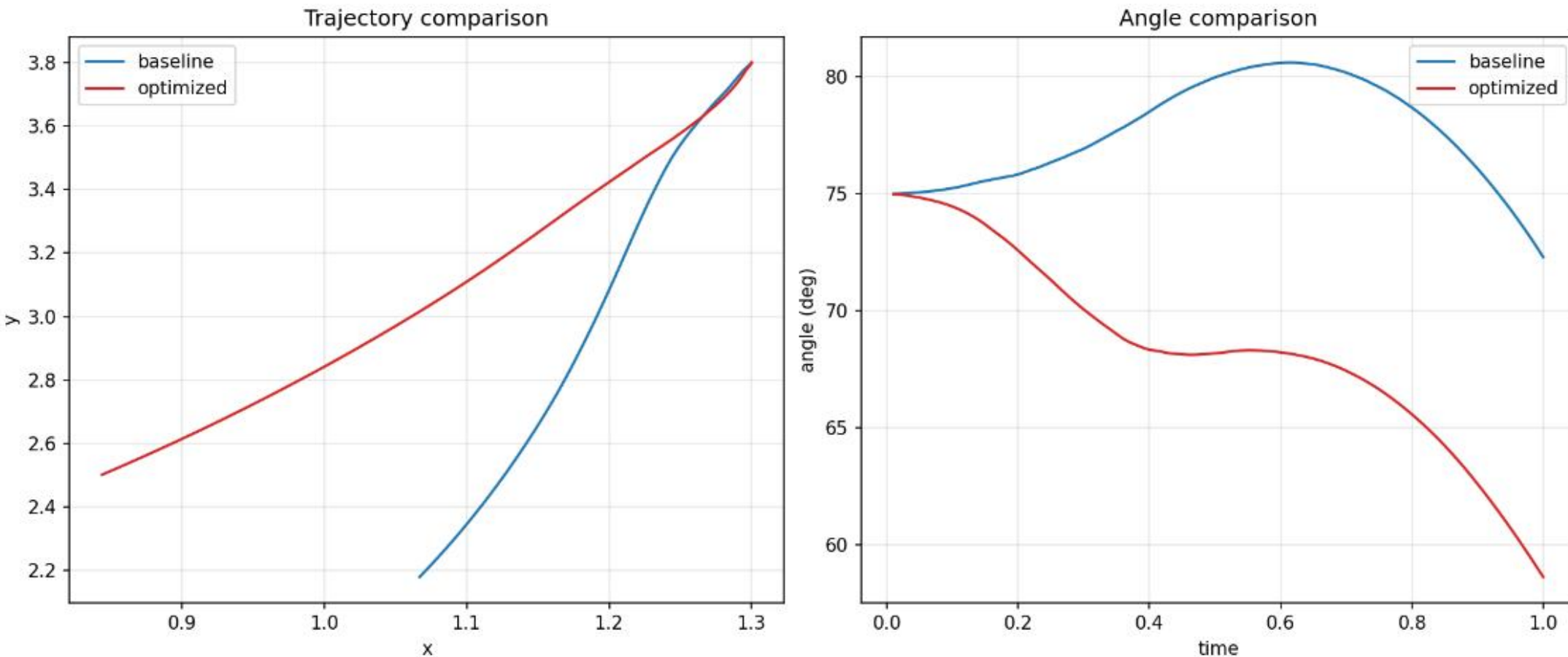


Figure 3 Trajectory and orientation comparisons between baseline and optimized CoM locations under $U_{wind} = 0.01$ and aspect ratio $\alpha = 7.0$

Conclusions

• Key Findings
The proposed framework captures diverse descent behaviors and enables stable optimization of center-of-mass location through differentiable simulation.

• Future Work
Develop truncated BPTT methods to reduce computational cost and support longer simulation horizons while maintaining physical consistency.

References

[1] A. Belmonte, H. Eisenberg, and E. Moses, “From Flutter to Tumble: Inertial Drag and Froude Similarity in Falling Paper,” Physical Review Letters, vol. 81, no. 2, pp. 345–348, 1998.

[2] U. Pesavento and Z. J. Wang, “Falling Paper: Navier-Stokes Solutions, Model of Fluid Forces, and Center of Mass Elevation,” Physical Review Letters, vol. 93, no. 14, Art. no. 144501, 2004.

[3] A. Andersen, U. Pesavento, and Z. J. Wang, “Unsteady Aerodynamics of Fluttering and Tumbling Plates,” Journal of Fluid Mechanics, vol. 541, pp. 65–90, 2005, doi: 10.1017/S002211200500594X.

[4] A. Andersen, U. Pesavento, and Z. J. Wang, “Analysis of Transitions between Fluttering, Tumbling and Steady Descent of Falling Cards,” Journal of Fluid Mechanics, vol. 541, pp. 91–104, 2005.

[5] E. M. Lau, W.-X. Huang, and C.-X. Xu, “Progression of Heavy Plates from Stable Falling to Tumbling Flight,” Physical Review Fluids, vol. 3, no. 7, Art. no. 074801, 2018.

[6] A. Rana, V. K. Kushwaha, and A. K. De, “A Numerical Study on the Effect of Aspect Ratio and Density Ratio on the Dynamics of Freely Falling Plate in the Flutter-to-Tumble Transition Regime,” Physics of Fluids, vol. 32, no. 7, Art. no. 073603, 2020.

[7] Q. Le, J. Bu, Y. Qu, B. Zhu, and T. Du, “Computational Biomimetics of Winged Seeds,” ACM Transactions on Graphics, vol. 43, no. 6, Art. no. 180, pp. 1–13, 2024, doi: 10.1145/3687899.